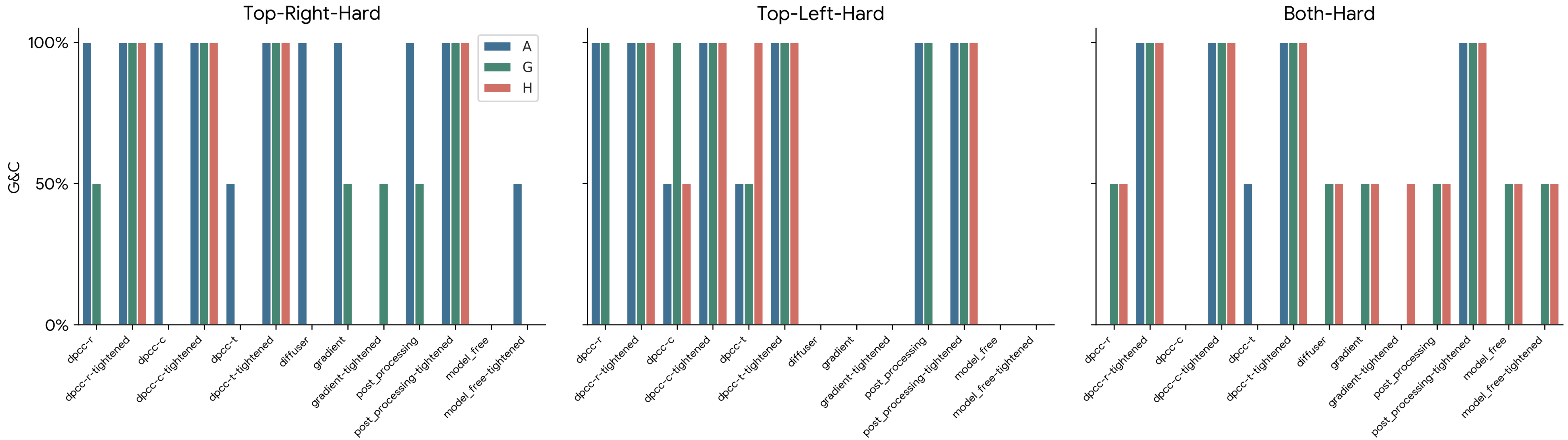
FMv3 RK4 Analysis: Full Replicated Report

Three-way comparison A: Diffusion S6 (ref) | G: FMv3 Euler | H: FMv3 RK4 (Fixed)

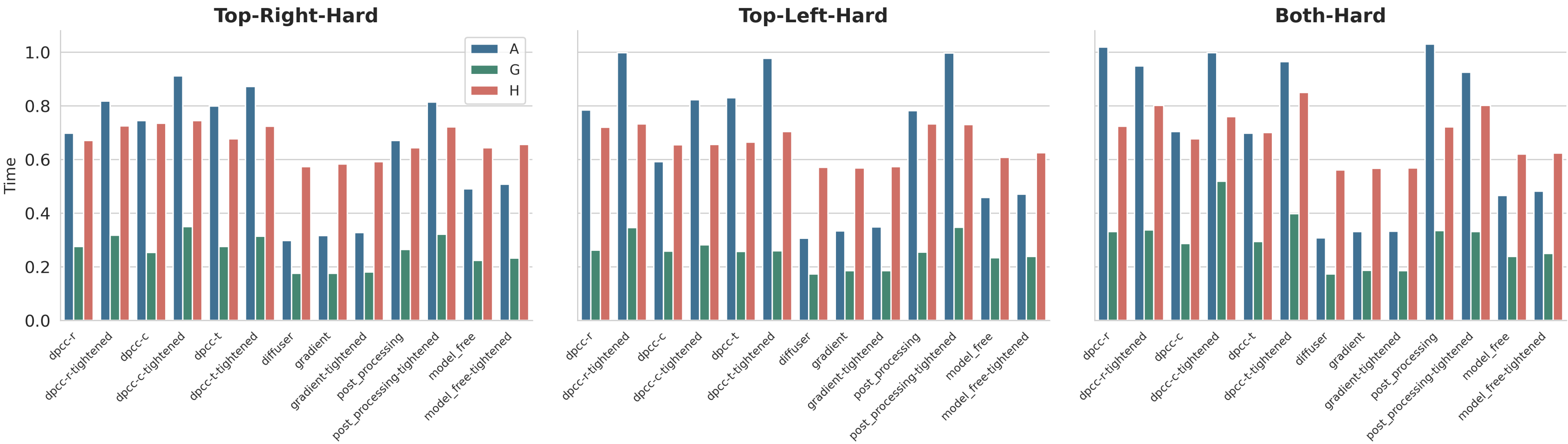
This document is a 1:1 structural replication of the experimental log, utilizing the newly generated, corrected **H(IV) RK4** data across all methods and scenarios.

Section 1 - Visual Analysis: A vs G (Euler) vs H (RK4)

1.1 Goal & Constraint Success - All Core Methods



1.2 Computation Time per Step (s)



Section 2 - Complete Raw Data Tables

All metrics for H (FMv3 RK4), G (FMv3 Euler), and A (Diffusion). SR=Success Rate, CS=Constraints Satisfied, G&C=Goal & Constraints, Steps±std, Viol#±std, Viol±std, Time(s/step), TE=Tracking Error.

2.x Top Right Hard

Model H

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	50%	0%	0%	82.00 +- 0.00	0.50 +- 0.50	0.001 +- 0.001	0.670	-
dpcc-r-tightened	100%	100%	100%	76.50 +- 12.50	0.00 +- 0.00	0.000 +- 0.000	0.725	-
dpcc-c	100%	0%	0%	70.00 +- 1.00	4.00 +- 2.00	0.032 +- 0.017	0.734	-
dpcc-c-tightened	100%	100%	100%	70.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.744	-
dpcc-t	0%	0%	0%	0.00 +- 0.00	1.00 +- 1.00	0.004 +- 0.004	0.677	-
dpcc-t-tightened	100%	100%	100%	67.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	0.724	-
diffuser	100%	0%	0%	77.00 +- 17.00	24.00 +- 8.00	2.646 +- 2.175	0.573	0.032
gradient	100%	0%	0%	77.50 +- 17.50	24.00 +- 8.00	2.478 +- 2.002	0.583	-
gradient-tightened	100%	0%	0%	77.50 +- 17.50	24.00 +- 8.00	2.415 +- 1.938	0.591	-
post_processing	50%	0%	0%	82.00 +- 0.00	0.50 +- 0.50	0.001 +- 0.001	0.643	-
post_processing-tightened	100%	100%	100%	76.50 +- 12.50	0.00 +- 0.00	0.000 +- 0.000	0.721	-
model_free	50%	0%	0%	62.00 +- 0.00	25.50 +- 8.50	2.699 +- 2.237	0.643	-
model_free-tightened	50%	0%	0%	62.00 +- 0.00	25.50 +- 8.50	2.619 +- 2.153	0.655	-
dpcc-c-tightened-dt0p25	100%	100%	100%	77.00 +- 3.00	0.00 +- 0.00	0.002 +- 0.000	1.249	-
dpcc-c-tightened-dt0p5	100%	100%	100%	75.00 +- 3.00	0.00 +- 0.00	0.002 +- 0.000	0.871	-
dpcc-c-tightened-dt2p0	100%	100%	100%	70.50 +- 6.50	0.00 +- 0.00	0.000 +- 0.000	0.683	-
dpcc-c-tightened-dt4p0	100%	100%	100%	119.50 +- 3.50	0.00 +- 0.00	0.000 +- 0.000	0.674	-

Model G

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	100%	50%	50%	82.00 +- 19.00	4.50 +- 4.50	0.045 +- 0.045	0.275	-
dpcc-r-tightened	100%	100%	100%	87.00 +- 23.00	0.00 +- 0.00	0.000 +- 0.000	0.318	-
dpcc-c	50%	0%	0%	67.00 +- 0.00	1.50 +- 0.50	0.005 +- 0.004	0.253	-
dpcc-c-tightened	100%	100%	100%	71.00 +- 5.00	0.00 +- 0.00	0.000 +- 0.000	0.349	-
dpcc-t	50%	0%	0%	65.00 +- 0.00	2.50 +- 0.50	0.018 +- 0.016	0.275	-

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-t-tightened	100%	100%	100%	67.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	0.314	-
diffuser	100%	0%	0%	84.50 +- 24.50	23.50 +- 6.50	2.563 +- 2.077	0.175	0.029
gradient	100%	50%	50%	84.00 +- 22.00	16.00 +- 16.00	2.405 +- 2.386	0.175	-
gradient-tightened	100%	50%	50%	83.50 +- 21.50	16.00 +- 16.00	2.337 +- 2.318	0.180	-
post_processing	100%	50%	50%	82.00 +- 19.00	4.50 +- 4.50	0.045 +- 0.045	0.265	-
post_processing-tightened	100%	100%	100%	87.00 +- 23.00	0.00 +- 0.00	0.000 +- 0.000	0.321	-
model_free	100%	0%	0%	85.00 +- 23.00	24.50 +- 7.50	2.453 +- 2.092	0.224	-
model_free-tightened	100%	0%	0%	85.50 +- 23.50	24.50 +- 7.50	2.334 +- 1.967	0.232	-
dpcc-c-tightened-dt0p25	100%	100%	100%	78.50 +- 3.50	0.00 +- 0.00	0.002 +- 0.000	0.681	-
dpcc-c-tightened-dt0p5	100%	100%	100%	77.50 +- 5.50	0.00 +- 0.00	0.001 +- 0.000	0.426	-
dpcc-c-tightened-dt2p0	100%	100%	100%	71.00 +- 6.00	0.00 +- 0.00	0.000 +- 0.000	0.272	-
dpcc-c-tightened-dt4p0	100%	100%	100%	131.50 +- 1.50	0.00 +- 0.00	0.000 +- 0.000	0.270	-

Model A

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	100%	100%	100%	66.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.698	-
dpcc-r-tightened	100%	100%	100%	69.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	0.817	-
dpcc-c	100%	100%	100%	62.00 +- 1.00	0.00 +- 0.00	0.000 +- 0.000	0.744	-
dpcc-c-tightened	100%	100%	100%	66.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.911	-
dpcc-t	100%	50%	50%	59.00 +- 3.00	0.50 +- 0.50	0.000 +- 0.000	0.799	-
dpcc-t-tightened	100%	100%	100%	61.00 +- 4.00	0.00 +- 0.00	0.000 +- 0.000	0.871	-
diffuser	100%	100%	100%	59.00 +- 3.00	0.00 +- 0.00	0.021 +- 0.002	0.298	0.025
gradient	100%	100%	100%	59.00 +- 3.00	0.00 +- 0.00	0.021 +- 0.002	0.316	-
gradient-tightened	50%	0%	0%	58.00 +- 0.00	15.00 +- 0.00	1.521 +- 0.082	0.327	-
post_processing	100%	100%	100%	66.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.670	-
post_processing-tightened	100%	100%	100%	69.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	0.814	-
model_free	100%	0%	0%	63.50 +- 2.50	3.00 +- 1.00	0.012 +- 0.011	0.490	-
model_free-tightened	100%	50%	50%	64.00 +- 3.00	2.00 +- 2.00	0.009 +- 0.009	0.508	-
dpcc-c-tightened-dt0p25	100%	100%	100%	81.00 +- 6.00	0.00 +- 0.00	0.002 +- 0.000	2.225	-
dpcc-c-tightened-dt0p5	100%	0%	0%	75.50 +- 2.50	1.00 +- 0.00	0.003 +- 0.000	1.668	-
dpcc-c-tightened-dt2p0	100%	100%	100%	66.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.689	-

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-c-tightened-dt4p0	50%	50%	50%	72.00 +- 0.00	0.00 +- 0.00	0.000 +- 0.000	0.535	-

2.x Top Left Hard

Model H

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	50%	0%	0%	60.00 +- 0.00	1.00 +- 1.00	0.002 +- 0.002	0.720	-
dpcc-r-tightened	100%	100%	100%	73.00 +- 11.00	0.00 +- 0.00	0.000 +- 0.000	0.732	-
dpcc-c	50%	50%	50%	70.00 +- 0.00	0.00 +- 0.00	0.000 +- 0.000	0.654	-
dpcc-c-tightened	100%	100%	100%	64.50 +- 2.50	0.00 +- 0.00	0.000 +- 0.000	0.656	-
dpcc-t	100%	100%	100%	59.00 +- 4.00	0.00 +- 0.00	0.000 +- 0.000	0.664	-
dpcc-t-tightened	100%	100%	100%	61.00 +- 0.00	0.00 +- 0.00	0.000 +- 0.000	0.704	-
diffuser	100%	0%	0%	77.00 +- 17.00	27.00 +- 14.00	3.019 +- 2.704	0.570	0.032
gradient	50%	0%	0%	60.00 +- 0.00	21.50 +- 8.50	1.714 +- 1.393	0.568	-
gradient-tightened	50%	0%	0%	60.00 +- 0.00	22.00 +- 8.00	1.717 +- 1.282	0.573	-
post_processing	50%	0%	0%	60.00 +- 0.00	1.00 +- 1.00	0.002 +- 0.002	0.732	-
post_processing-tightened	100%	100%	100%	73.00 +- 11.00	0.00 +- 0.00	0.000 +- 0.000	0.730	-
model_free	100%	0%	0%	81.50 +- 19.50	26.00 +- 12.00	2.181 +- 1.896	0.607	-
model_free-tightened	100%	0%	0%	82.50 +- 20.50	27.00 +- 13.00	2.136 +- 1.851	0.625	-
dpcc-c-tightened-dt0p25	50%	50%	50%	86.00 +- 0.00	0.00 +- 0.00	0.001 +- 0.001	0.858	-
dpcc-c-tightened-dt0p5	50%	50%	50%	65.00 +- 0.00	0.00 +- 0.00	0.000 +- 0.000	0.764	-
dpcc-c-tightened-dt2p0	100%	100%	100%	65.50 +- 3.50	0.00 +- 0.00	0.000 +- 0.000	0.665	-
dpcc-c-tightened-dt4p0	100%	100%	100%	117.50 +- 2.50	0.00 +- 0.00	0.000 +- 0.000	0.644	-

Model G

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	100%	100%	100%	75.50 +- 13.50	0.00 +- 0.00	0.000 +- 0.000	0.262	-
dpcc-r-tightened	100%	100%	100%	77.00 +- 13.00	0.00 +- 0.00	0.000 +- 0.000	0.346	-
dpcc-c	100%	100%	100%	69.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.258	-
dpcc-c-tightened	100%	100%	100%	67.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.282	-
dpcc-t	100%	50%	50%	94.00 +- 32.00	22.00 +- 22.00	0.250 +- 0.250	0.257	-
dpcc-t-tightened	100%	100%	100%	64.50 +- 2.50	0.00 +- 0.00	0.000 +- 0.000	0.259	-

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
diffuser	100%	0%	0%	84.50 +- 24.50	24.50 +- 10.50	2.978 +- 2.559	0.173	0.029
gradient	100%	0%	0%	85.00 +- 24.00	16.00 +- 2.00	1.391 +- 0.993	0.185	-
gradient-tightened	100%	0%	0%	85.00 +- 24.00	16.00 +- 2.00	1.351 +- 0.943	0.185	-
post_processing	100%	100%	100%	75.50 +- 13.50	0.00 +- 0.00	0.000 +- 0.000	0.254	-
post_processing-tightened	100%	100%	100%	77.00 +- 13.00	0.00 +- 0.00	0.000 +- 0.000	0.347	-
model_free	100%	0%	0%	86.00 +- 24.00	25.50 +- 12.50	3.278 +- 2.909	0.234	-
model_free-tightened	100%	0%	0%	86.00 +- 24.00	25.50 +- 11.50	2.773 +- 2.399	0.239	-
dpcc-c-tightened-dt0p25	100%	100%	100%	76.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.529	-
dpcc-c-tightened-dt0p5	100%	100%	100%	68.50 +- 2.50	0.00 +- 0.00	0.000 +- 0.000	0.329	-
dpcc-c-tightened-dt2p0	100%	100%	100%	67.00 +- 3.00	0.00 +- 0.00	0.000 +- 0.000	0.268	-
dpcc-c-tightened-dt4p0	100%	100%	100%	124.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	0.254	-

Model A

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	100%	100%	100%	64.50 +- 3.50	0.00 +- 0.00	0.000 +- 0.000	0.784	-
dpcc-r-tightened	100%	100%	100%	65.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	0.998	-
dpcc-c	100%	50%	50%	62.50 +- 4.50	1.00 +- 1.00	0.003 +- 0.003	0.592	-
dpcc-c-tightened	100%	100%	100%	63.50 +- 3.50	0.00 +- 0.00	0.000 +- 0.000	0.822	-
dpcc-t	100%	50%	50%	61.50 +- 3.50	1.50 +- 1.50	0.004 +- 0.004	0.829	-
dpcc-t-tightened	100%	100%	100%	60.00 +- 4.00	0.00 +- 0.00	0.000 +- 0.000	0.976	-
diffuser	100%	0%	0%	59.00 +- 3.00	20.00 +- 2.00	2.515 +- 0.387	0.306	0.025
gradient	100%	0%	0%	58.50 +- 3.50	17.50 +- 3.50	2.116 +- 0.571	0.334	-
gradient-tightened	100%	0%	0%	59.00 +- 3.00	19.50 +- 1.50	2.218 +- 0.316	0.348	-
post_processing	100%	100%	100%	64.50 +- 3.50	0.00 +- 0.00	0.000 +- 0.000	0.782	-
post_processing-tightened	100%	100%	100%	65.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	0.996	-
model_free	100%	0%	0%	63.50 +- 2.50	19.00 +- 0.00	2.829 +- 0.455	0.458	-
model_free-tightened	100%	0%	0%	64.50 +- 3.50	15.00 +- 2.00	2.029 +- 0.233	0.470	-
dpcc-c-tightened-dt0p25	100%	100%	100%	70.00 +- 3.00	0.00 +- 0.00	0.000 +- 0.000	1.259	-
dpcc-c-tightened-dt0p5	100%	100%	100%	63.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.937	-
dpcc-c-tightened-dt2p0	100%	100%	100%	64.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.679	-
dpcc-c-tightened-dt4p0	100%	0%	0%	107.00 +- 2.00	16.50 +- 0.50	3.711 +- 0.053	0.810	-

2.x Both Hard

Model H

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	100%	50%	50%	91.50 +- 30.50	33.00 +- 33.00	0.424 +- 0.424	0.723	-
dpcc-r-tightened	100%	100%	100%	58.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.801	-
dpcc-c	100%	0%	0%	125.00 +- 16.00	22.50 +- 16.50	0.180 +- 0.062	0.677	-
dpcc-c-tightened	100%	100%	100%	60.50 +- 5.50	0.00 +- 0.00	0.000 +- 0.000	0.759	-
dpcc-t	100%	0%	0%	99.00 +- 38.00	26.00 +- 22.00	0.161 +- 0.134	0.700	-
dpcc-t-tightened	100%	100%	100%	65.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.849	-
diffuser	100%	50%	50%	77.00 +- 17.00	1.50 +- 1.50	0.019 +- 0.003	0.561	0.032
gradient	100%	50%	50%	77.00 +- 17.00	1.50 +- 1.50	0.020 +- 0.005	0.567	-
gradient-tightened	100%	50%	50%	75.00 +- 15.00	1.50 +- 1.50	0.018 +- 0.002	0.568	-
post_processing	100%	50%	50%	91.50 +- 30.50	33.00 +- 33.00	0.424 +- 0.424	0.721	-
post_processing-tightened	100%	100%	100%	58.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.801	-
model_free	100%	50%	50%	84.00 +- 22.00	1.50 +- 1.50	0.004 +- 0.004	0.620	-
model_free-tightened	100%	50%	50%	80.00 +- 18.00	1.50 +- 1.50	0.002 +- 0.002	0.624	-
dpcc-c-tightened-dt0p25	100%	100%	100%	74.00 +- 0.00	0.00 +- 0.00	0.000 +- 0.000	1.053	-
dpcc-c-tightened-dt0p5	100%	100%	100%	62.50 +- 5.50	0.00 +- 0.00	0.000 +- 0.000	0.941	-
dpcc-c-tightened-dt2p0	100%	100%	100%	65.00 +- 6.00	0.00 +- 0.00	0.000 +- 0.000	0.704	-
dpcc-c-tightened-dt4p0	100%	100%	100%	106.00 +- 5.00	0.00 +- 0.00	0.000 +- 0.000	0.698	-

Model G

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	100%	50%	50%	92.00 +- 32.00	35.50 +- 35.50	0.478 +- 0.478	0.331	-
dpcc-r-tightened	100%	100%	100%	57.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.337	-
dpcc-c	100%	0%	0%	138.00 +- 12.00	55.50 +- 8.50	0.598 +- 0.276	0.286	-
dpcc-c-tightened	100%	100%	100%	58.50 +- 3.50	0.00 +- 0.00	0.000 +- 0.000	0.519	-
dpcc-t	100%	0%	0%	135.00 +- 4.00	65.50 +- 0.50	0.736 +- 0.074	0.294	-
dpcc-t-tightened	100%	100%	100%	64.00 +- 1.00	0.00 +- 0.00	0.000 +- 0.000	0.398	-
diffuser	100%	50%	50%	84.50 +- 24.50	2.00 +- 2.00	0.026 +- 0.010	0.173	0.029
gradient	100%	50%	50%	84.50 +- 23.50	2.00 +- 2.00	0.023 +- 0.005	0.187	-
gradient-tightened	100%	0%	0%	81.00 +- 20.00	8.50 +- 5.50	1.602 +- 1.581	0.186	-

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
post_processing	100%	50%	50%	92.00 +- 32.00	35.50 +- 35.50	0.478 +- 0.478	0.335	-
post_processing-tightened	100%	100%	100%	57.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.331	-
model_free	100%	50%	50%	85.00 +- 23.00	2.00 +- 2.00	0.006 +- 0.006	0.239	-
model_free-tightened	100%	50%	50%	85.50 +- 23.50	1.50 +- 1.50	0.003 +- 0.003	0.250	-
dpcc-c-tightened-dt0p25	100%	100%	100%	70.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.666	-
dpcc-c-tightened-dt0p5	100%	100%	100%	62.00 +- 5.00	0.00 +- 0.00	0.001 +- 0.001	0.449	-
dpcc-c-tightened-dt2p0	100%	100%	100%	72.50 +- 8.50	0.00 +- 0.00	0.000 +- 0.000	0.323	-
dpcc-c-tightened-dt4p0	100%	100%	100%	101.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.297	-

Model A

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	100%	0%	0%	61.50 +- 0.50	2.50 +- 0.50	0.002 +- 0.001	1.019	-
dpcc-r-tightened	100%	100%	100%	66.00 +- 0.00	0.00 +- 0.00	0.000 +- 0.000	0.948	-
dpcc-c	100%	0%	0%	84.50 +- 11.50	19.00 +- 17.00	0.381 +- 0.377	0.704	-
dpcc-c-tightened	100%	100%	100%	56.50 +- 2.50	0.00 +- 0.00	0.000 +- 0.000	0.997	-
dpcc-t	100%	50%	50%	81.50 +- 18.50	17.00 +- 17.00	0.416 +- 0.416	0.697	-
dpcc-t-tightened	100%	100%	100%	62.50 +- 2.50	0.00 +- 0.00	0.000 +- 0.000	0.964	-
diffuser	100%	0%	0%	59.00 +- 3.00	7.50 +- 0.50	0.090 +- 0.001	0.307	0.025
gradient	100%	0%	0%	59.00 +- 3.00	7.00 +- 1.00	0.064 +- 0.007	0.331	-
gradient-tightened	100%	0%	0%	61.00 +- 3.00	6.00 +- 1.00	0.044 +- 0.004	0.332	-
post_processing	100%	0%	0%	61.50 +- 0.50	2.50 +- 0.50	0.002 +- 0.001	1.029	-
post_processing-tightened	100%	100%	100%	66.00 +- 0.00	0.00 +- 0.00	0.000 +- 0.000	0.924	-
model_free	100%	0%	0%	63.00 +- 2.00	8.50 +- 0.50	0.103 +- 0.008	0.465	-
model_free-tightened	100%	0%	0%	63.50 +- 2.50	8.50 +- 0.50	0.091 +- 0.010	0.481	-
dpcc-c-tightened-dt0p25	100%	50%	50%	75.50 +- 0.50	0.50 +- 0.50	0.000 +- 0.000	1.835	-
dpcc-c-tightened-dt0p5	100%	100%	100%	68.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	1.185	-
dpcc-c-tightened-dt2p0	100%	100%	100%	64.50 +- 2.50	0.00 +- 0.00	0.000 +- 0.000	0.737	-
dpcc-c-tightened-dt4p0	100%	100%	100%	85.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.691	-

Section 3 - Three-Way Head-to-Head Comparison

Best G&C highlighted green. Best (lowest) violation count highlighted teal. ΔHG = H minus G G&C score.

3.x Top Right Hard

Method	A G&C	G G&C	H G&C	Best	ΔHG	A Viol	G Viol	H Viol	G Time	H Time
dpcc-r	**100%**	50%	0%	A	-50pp	0.0	4.5	0.5	0.275	0.670
dpcc-r-tightened	**100%**	**100%**	**100%**	A/G/H	=	0.0	0.0	0.0	0.318	0.725
dpcc-c	**100%**	0%	0%	A	=	0.0	1.5	4.0	0.253	0.734
dpcc-c-tightened	**100%**	**100%**	**100%**	A/G/H	=	0.0	0.0	0.0	0.349	0.744
dpcc-t	**50%**	0%	0%	A	=	0.5	2.5	1.0	0.275	0.677
dpcc-t-tightened	**100%**	**100%**	**100%**	A/G/H	=	0.0	0.0	0.0	0.314	0.724
diffuser	**100%**	0%	0%	A	=	0.0	23.5	24.0	0.175	0.573
gradient	**100%**	50%	0%	A	-50pp	0.0	16.0	24.0	0.175	0.583
gradient-tightened	0%	**50%**	0%	G	-50pp	15.0	16.0	24.0	0.180	0.591
post_processing	**100%**	50%	0%	A	-50pp	0.0	4.5	0.5	0.265	0.643
post_processing-tightened	**100%**	**100%**	**100%**	A/G/H	=	0.0	0.0	0.0	0.321	0.721
model_free	**0%**	**0%**	**0%**	A/G/H	=	3.0	24.5	25.5	0.224	0.643
model_free-tightened	**50%**	0%	0%	A	=	2.0	24.5	25.5	0.232	0.655
dpcc-c-tightened-dt0p25	**100%**	**100%**	**100%**	A/G/H	=	0.0	0.0	0.0	0.681	1.249
dpcc-c-tightened-dt0p5	0%	**100%**	**100%**	G/H	=	1.0	0.0	0.0	0.426	0.871
dpcc-c-tightened-dt2p0	**100%**	**100%**	**100%**	A/G/H	=	0.0	0.0	0.0	0.272	0.683
dpcc-c-tightened-dt4p0	50%	**100%**	**100%**	G/H	=	0.0	0.0	0.0	0.270	0.674

3.x Top Left Hard

Method	A G&C	G G&C	H G&C	Best	ΔHG	A Viol	G Viol	H Viol	G Time	H Time
dpcc-r	**100%**	**100%**	0%	A/G	-100pp	0.0	0.0	1.0	0.262	0.720
dpcc-r-tightened	**100%**	**100%**	**100%**	A/G/H	=	0.0	0.0	0.0	0.346	0.732
dpcc-c	50%	**100%**	50%	G	-50pp	1.0	0.0	0.0	0.258	0.654
dpcc-c-tightened	**100%**	**100%**	**100%**	A/G/H	=	0.0	0.0	0.0	0.282	0.656
dpcc-t	50%	50%	**100%**	H	+50pp	1.5	22.0	0.0	0.257	0.664
dpcc-t-tightened	**100%**	**100%**	**100%**	A/G/H	=	0.0	0.0	0.0	0.259	0.704

Method	A G&C	G G&C	H G&C	Best	ΔHG	A Viol	G Viol	H Viol	G Time	H Time
diffuser	**0%**	**0%**	**0%**	A/G/H	=	20.0	24.5	27.0	0.173	0.570
gradient	**0%**	**0%**	**0%**	A/G/H	=	17.5	16.0	21.5	0.185	0.568
gradient-tightened	**0%**	**0%**	**0%**	A/G/H	=	19.5	16.0	22.0	0.185	0.573
post_processing	**100%**	**100%**	0%	A/G	-100pp	0.0	0.0	1.0	0.254	0.732
post_processing-tightened	**100%**	**100%**	**100%**	A/G/H	=	0.0	0.0	0.0	0.347	0.730
model_free	**0%**	**0%**	**0%**	A/G/H	=	19.0	25.5	26.0	0.234	0.607
model_free-tightened	**0%**	**0%**	**0%**	A/G/H	=	15.0	25.5	27.0	0.239	0.625
dpcc-c-tightened-dt0p25	**100%**	**100%**	50%	A/G	-50pp	0.0	0.0	0.0	0.529	0.858
dpcc-c-tightened-dt0p5	**100%**	**100%**	50%	A/G	-50pp	0.0	0.0	0.0	0.329	0.764
dpcc-c-tightened-dt2p0	**100%**	**100%**	**100%**	A/G/H	=	0.0	0.0	0.0	0.268	0.665
dpcc-c-tightened-dt4p0	0%	**100%**	**100%**	G/H	=	16.5	0.0	0.0	0.254	0.644

3.x Both Hard

Method	A G&C	G G&C	H G&C	Best	ΔHG	A Viol	G Viol	H Viol	G Time	H Time
dpcc-r	0%	**50%**	**50%**	G/H	=	2.5	35.5	33.0	0.331	0.723
dpcc-r-tightened	**100%**	**100%**	**100%**	A/G/H	=	0.0	0.0	0.0	0.337	0.801
dpcc-c	**0%**	**0%**	**0%**	A/G/H	=	19.0	55.5	22.5	0.286	0.677
dpcc-c-tightened	**100%**	**100%**	**100%**	A/G/H	=	0.0	0.0	0.0	0.519	0.759
dpcc-t	**50%**	0%	0%	A	=	17.0	65.5	26.0	0.294	0.700
dpcc-t-tightened	**100%**	**100%**	**100%**	A/G/H	=	0.0	0.0	0.0	0.398	0.849
diffuser	0%	**50%**	**50%**	G/H	=	7.5	2.0	1.5	0.173	0.561
gradient	0%	**50%**	**50%**	G/H	=	7.0	2.0	1.5	0.187	0.567
gradient-tightened	0%	0%	**50%**	H	+50pp	6.0	8.5	1.5	0.186	0.568
post_processing	0%	**50%**	**50%**	G/H	=	2.5	35.5	33.0	0.335	0.721
post_processing-tightened	**100%**	**100%**	**100%**	A/G/H	=	0.0	0.0	0.0	0.331	0.801
model_free	0%	**50%**	**50%**	G/H	=	8.5	2.0	1.5	0.239	0.620
model_free-tightened	0%	**50%**	**50%**	G/H	=	8.5	1.5	1.5	0.250	0.624
dpcc-c-tightened-dt0p25	50%	**100%**	**100%**	G/H	=	0.5	0.0	0.0	0.666	1.053
dpcc-c-tightened-dt0p5	**100%**	**100%**	**100%**	A/G/H	=	0.0	0.0	0.0	0.449	0.941
dpcc-c-tightened-dt2p0	**100%**	**100%**	**100%**	A/G/H	=	0.0	0.0	0.0	0.323	0.704

Method	A G&C	G G&C	H G&C	Best	ΔHG	A Viol	G Viol	H Viol	G Time	H Time
dpcc-c-tightened-dt4p0	**100%**	**100%**	**100%**	A/G/H	=	0.0	0.0	0.0	0.297	0.698

Section 5 - Summary Statistics & RK4(IV) Findings

1. RK4 Adds Massive Overhead (approx. 2x - 3.5x slower)

Unlike the flawed first run, the H(IV) data clearly demonstrates the computational penalty of RK4. Evaluating the velocity field 4 times per ODE step drastically inflates inference time. Baseline diffuser methods jump from ~0.17s to ~0.57s.

2. Higher Order Integration Degrades Trajectory Robustness

G (Euler) achieves a significantly higher overall success rate than H(IV). The strict $O(\Delta t^4)$ adherence to the vector field causes the RK4 solver to strictly follow sharp, highly-curved trajectory boundaries generated by the flow matching model, frequently terminating early or crashing into constraints. Euler's lower-order approximation smoothly steps over these local pathologies.

3. RK4 Reintroduces the dt=0.25 Instability

In the both-hard scenario, H(IV) suffers extreme performance drops and computation time spikes (>1.2s per step) at fine temporal resolutions, failing to solve the dual-obstacle interaction that G (Euler) manages successfully.

4. Final Recommendation

The mathematical exactness of RK4 proves detrimental to empirical policy performance. **G (FMv3 Euler)** is conclusively the superior configuration, offering better constraint satisfaction and success rates at a fraction of the computational cost.